6G Weekly Survey — Oct 05 - Oct 11, 2026

15 items across 4 topics, ranked by importance.

6G Physical Layer

Multi-feed Plane Wave Generator for Compact OTA Multi-Target Emulation With Joint Angle-Delay-Doppler Control

Le Yu, Guangming Dong, Chunhui Li, Wei Fan · arXiv eess.SP · 2026-09-30

★★★★■ Addresses critical ISAC testing infrastructure gap by enabling practical lab validation of multi-target sensing without massive anechoic chambers, directly supporting 6G ISAC device development with validated sub-far-field test methodology.

This paper proposes a multi-feed plane wave generator (PWG) for compact over-the-air (OTA) testing of ISAC radar capabilities, enabling simultaneous emulation of spatially close targets with independent angle, range (delay), and Doppler control at substantially reduced test distances. The system uses a multi-input amplitude-phase matrix to map target signals to PWG excitation vectors that synthesize local plane waves, validated experimentally at 3 GHz with a 1×11 phased array at 1.6 m separation (versus 7.2 m far-field requirement). Results show ≤1° angular error, ≤2.4 ns delay error, and ≤0.15 m/s velocity error across multi-target scenarios.

Contribution: First demonstration of compact-range multi-target radar emulation with joint angle-delay-Doppler control at distances far shorter than Fraunhofer, using multi-feed PWG synthesis to create independent local plane waves for each target direction simultaneously.

Relevance to your work: Directly enables experimental validation of ISAC systems you research—this compact multi-target OTA framework would allow realistic testing of joint sensing-communication PHY designs, radar signal processing algorithms, and beamforming schemes for ISAC without requiring far-field facilities.

Low Overhead IMU Assisted Predictive Beam Management for Multiband LEO Direct to Device Links

Abdulrahman Al Hababi, Meysam Ghanbari, Mohammad Taghi Dabiri, Mazen O. Hasna, et al. · arXiv eess.SP · 2026-10-01

★★★★■ Directly addresses a critical NTN challenge (sub-RTT beam tracking for mobile D2D) with a deployable solution validated on realistic motion traces, highly relevant as Ku/Ka-band LEO D2D enters commercial trials.

This paper addresses beam management for LEO direct-to-device connectivity at Ku-band, where handset motion disrupts alignment within milliseconds across hundreds of satellite-panel-beam candidates. The authors leverage IMU data to predict current orientation from delayed measurements, analytically pre-score beams using satellite ephemeris and panel geometry, then train only a pruned candidate set under fixed pilot budget, activating Ku-band only when predicted rate exceeds a low-frequency anchor link. Trace-driven simulations with four-panel handsets show 21% goodput gain and 58% outage reduction versus delayed-attitude baseline at 0.3% overhead, operating within 4% of a zero-overhead oracle.

Contribution: First practical beam management framework fusing low-latency IMU prediction with geometric beam scoring to handle fast handset rotation in multiband LEO direct-to-device systems under strict pilot budget constraints.

Relevance to your work: Highly relevant to your NTN beam management work—demonstrates practical sensor fusion and predictive beam selection for fast-moving LEO terminals, complementing ML-based beam prediction methods and offering a low-overhead baseline for future AI-native NTN physical layers.

Energy-Driven Evaluation of Network Digital Twinning Applied to mmWave Beam Management

João Borges, Bruno Castro, Kleber Cardoso, Andrey Silva, et al. · arXiv cs.NI · 2026-09-28

★★★★■ Strong practical contribution bridging NDT theory with real mmWave hardware and energy-aware optimization, directly addressing 6G operational challenges with credible experimental validation using Sionna and custom 60 GHz testbed.

This paper evaluates Network Digital Twins (NDTs) for mmWave beam management by jointly optimizing twinning rate and fidelity to minimize energy consumption, using real 60 GHz hardware measurements instead of pure simulation. The physical twin uses custom Wi-Fi radios with 32 antenna elements, while the virtual twin employs Sionna ray tracing; Bayesian Optimization is applied to tune operational parameters. Results show that fidelity levels fluctuate during operation and that sample-efficient methods can optimize twinning rates even at reduced fidelity.

Contribution: First rigorous study examining the coupled impact of twinning rate and fidelity on NDT energy consumption using hardware-measured physical twins rather than simulated data, demonstrating that Bayesian Optimization can efficiently tune these parameters for real mmWave beam management.

Relevance to your work: Highly relevant to your beam management and massive MIMO interests, particularly the intersection of learned models (ray tracing), CSI optimization, and practical deployment constraints for AI-native air interfaces in 6G.

Pinching-Antenna Differential Spatial Modulation with RSSI-Assisted Selection and Low-Complexity Detection: Methods and Performance Analysis

Yusuf Akar, Mahmoud Raeisi, Henk Wymeersch, Mikko Valkama, et al. · arXiv eess.SP · 2026-09-29

★★★■■ Solid incremental work addressing CSI overhead for reconfigurable antennas in 6G FR3 bands, but pinching-antenna technology is still immature and differential modulation has limited spectral efficiency compared to coherent AI-native approaches.

The paper proposes pinching-antenna differential spatial modulation (PA-DSM) that enables data transmission without requiring instantaneous CSI at the receiver by using slow-timescale RSSI measurements to select favorable antenna positions along a reconfigurable dielectric waveguide. A closed-form low-complexity ML detector is derived for PSK modulation, and pairwise error probability bounds are developed under normalized common-K Rician fading to characterize performance. Validation across 6G FR3 scenarios shows PA-DSM can outperform coherent schemes when channel estimates become outdated between pilots in high-mobility environments.

Contribution: First application of differential spatial modulation to pinching-antenna systems with RSSI-based slow-timescale PA selection that eliminates instantaneous CSI requirements while exploiting macroscopic spatial reconfigurability.

Relevance to your work: Relevant to your physical-layer research as it addresses CSI acquisition overhead in novel antenna architectures for 6G, though the non-coherent differential approach contrasts with neural receiver designs that could jointly learn channel estimation and detection.

6G Demos & Trials

Agentic RF Intelligence: Multi-Timescale 6G Sensing and Reasoning with On-Device Foundation Models

Jaron Fontaine, Jelle De Moerloose, Xander Vanparys, Anton Lambrecht, et al. · arXiv eess.SP · 2026-10-02

★★★★■ Addresses critical 6G edge intelligence challenge with working prototype combining two emerging paradigms (PHY foundation models and LLM agents), though operational scope remains limited to sensing rather than full network control.

This paper proposes a dual-loop architecture for 6G network edge devices that decouples fast physical-layer sensing (millisecond-scale spectrum monitoring, interference detection via wireless PHY foundation models) from slower agentic reasoning (second-scale LLM-based orchestration and policy adaptation). The authors implement an on-device prototype on NVIDIA Jetson Thor with USRP B200 Mini, demonstrating real-time IQ sample processing with WPFMs running independently while an LLM asynchronously interprets RF events and steers sensing actions. Experiments confirm the timescale separation enables continuous fast-loop

operation during agentic reasoning cycles.

Contribution: First demonstrated integration of wireless physical-layer foundation models with LLM-based agentic reasoning in a single edge device prototype, establishing feasibility of multi-timescale intelligence for autonomous 6G network control.

Relevance to your work: Directly relevant: demonstrates practical deployment of AI-native physical layer (WPFMs for spectrum sensing) integrated with higher-layer reasoning, offering architectural blueprint for autonomous AI-native air interface operation and real-time PHY adaptation you are researching.

Beyond Idealized Orbits: Realistic Network Topology and Spatio-Temporal Traffic Emulation for LEO Satellite Constellations

Suryansh Aryan, Josh Smith, Alexa Houck, Rahul V. Chintalapati, et al. · arXiv cs.NI · 2026-10-02

★★★★■ Provides essential infrastructure for NTN research with validated realism against operational Starlink data, enabling credible protocol and architecture development for 6G non-terrestrial components without requiring actual satellite access.

This paper extends the SpaceNet testbed to emulate multi-shell LEO satellite constellations with realistic network topology, congestion modeling, and routing between space and terrestrial terminals. The authors validate their emulator against real Starlink ping and iPerf measurements, demonstrating accurate prediction of RTT and throughput temporal behaviors across different traffic loads. The testbed supports flexible design of inter-shell routing and congestion-aware load balancing for realistic NTN research.

Contribution: First open testbed enabling high-fidelity emulation of multi-shell LEO constellations with validated congestion modeling and routing that matches real Starlink network performance within 95% confidence intervals.

Relevance to your work: Directly supports your NTN research by providing a validated testbed for evaluating AI-based routing, resource allocation, and PHY-layer schemes in realistic multi-shell LEO scenarios before live deployment or field trials.

TSGuard: A Real-Time Framework for Detecting and Imputing Missing Data in Streaming Time Series

Imane Hocine, Asma Abboura, Soror Sahri, Abhijith Senthilkumar, et al. · arXiv cs.LG · 2026-10-02

★★■■■ Competent system engineering for sensor data quality but addresses a generic time-series problem rather than advancing wireless PHY or 6G-specific challenges; the graph-aware temporal model itself appears to be application of existing techniques.

TSGuard is a real-time system for detecting and imputing missing values in streaming sensor time series, combining a lightweight graph-aware temporal imputation model with constraint-based validation and operator-facing explanations. Unlike offline methods, it operates on live data streams and enforces domain plausibility through a validation loop that either retains original values as plausible anomalies or replaces constraint-violating observations. The demo targets environmental sensing applications where communication losses and sensor faults cause delayed or missing observations.

Contribution: The novel integration of lightweight spatiotemporal imputation with domain-aware constraint validation in a real-time streaming context, providing explicit retain-or-replace decisions with operator-level explainability rather than treating imputation as isolated prediction.

Relevance to your work: Tangential—while streaming sensor data quality matters for ISAC and NTN sensor networks, this work focuses on general time-series imputation rather than wireless channel estimation, CSI compression, or air interface design that directly informs 6G research.

Coverage You Can Steer: Online Conformal Calibration for RL-Driven Hardware-Aware NAS

Pedro Brandimarte, Nerea Aranjuelo, Marcos Nieto, Oihana Otaegui · arXiv cs.LG · 2026-10-02

★★■■■ Competent optimization work for AutoML efficiency, but addresses NAS for general deep learning rather than wireless-specific challenges like hardware-aware PHY design or on-device learning for 6G.

The paper addresses the computational bottleneck in hardware-aware neural architecture search (NAS) where each candidate architecture requires full training for evaluation. It applies online conformal prediction with adaptive coverage control to prune low-reward candidates during RL-driven NAS, maintaining statistical guarantees despite non-stationary proposal distributions. Experiments show 25-50% reduction in architecture evaluations while preserving search accuracy across multiple network families and search strategies.

Contribution: Introduces online feedback control (Adaptive Conformal Inference variants) to maintain distribution-free coverage guarantees in non-exchangeable RL-NAS settings, replacing static calibration that fails when the policy's architecture proposals evolve during search.

Relevance to your work: Tangential—the adaptive conformal framework could potentially transfer to neural receiver NAS under hardware constraints (e.g., selecting lightweight CSI compressor architectures), but the paper does not engage with wireless physical layer or 6G air interface problems.

HyperBrowseComp: A Multilingual and Multimodal Stress Test for Web-Browsing Agents

Alham Fikri Aji, Faiz Rizki Ramadhan, Zayd M. K. Zuhri, Seung Hun Eddie Han, et al. · arXiv cs.LG · 2026-10-02

★■■■■ While technically competent, this work addresses web-browsing agents and information retrieval, which is completely orthogonal to wireless communications, physical layer design, channel modeling, or any aspect of 6G research.

This paper presents a multilingual, multimodal web-browsing benchmark with 423 difficult questions across 13 languages requiring agents to locate obscure evidence through multi-step reasoning and heterogeneous sources (videos, images, maps). Questions are filtered to exclude those answerable without internet access, and both AI models and humans are evaluated on the task. The benchmark tests persistent information-seeking capabilities across languages and modalities on the open web.

Contribution: A rigorously constructed multilingual benchmark specifically designed to stress-test web-browsing agents on challenging multi-step, multi-modal information retrieval tasks that cannot be solved with parametric knowledge alone.

Relevance to your work: tangential

The Neuro-Physical Inverter: A Modular Framework for Magnetotelluric Inversion Coupling Ensemble Conditioning with Residual Learning

Jae Deok Kim, Sai Ravela, Rob. L. Evans · arXiv cs.LG · 2026-10-02

★■■■■ While methodologically sound, this work addresses magnetotelluric inversion in geophysics with no direct connection to wireless communications, PHY layer design, or 6G system problems.

The paper proposes a two-stage geophysical inversion framework (NPI) that first conditions a prior ensemble of resistivity models using an Ensemble-Conditional Gaussian Process, then applies a residual-learning neural network to refine predictions while propagating uncertainty. Demonstrated on 1D magnetotelluric data from Nevada, the method reduces misfit while maintaining ensemble spread for uncertainty quantification. The modular architecture is designed to be dimension-agnostic and scalable to higher-dimensional inverse problems.

Contribution: Novel coupling of ensemble conditioning with physics-informed residual learning for uncertainty-aware geophysical inversion, where uncertainty propagates through both stages rather than being discarded after initial estimation.

Relevance to your work: Tangential—the ensemble conditioning and residual learning principles could theoretically inspire inverse problems in channel estimation or environmental sensing, but the geophysical application domain and physics are fundamentally disconnected from wireless research.

AI-native air interface

Agentic Wireless Digital Twin Construction and Calibration Using Real-World Measurements

Fan-Hao Lin, Kun-Li Shih, Chu-Hsiang Huang, Hui Chen, et al. · arXiv cs.IT · 2026-10-01

★★★★■ Addresses a critical bottleneck in AI-native RAN development by automating digital twin creation with real-world validation, from a strong research group known for channel modeling work.

This paper introduces an agentic AI framework (AWDT) that autonomously constructs and calibrates wireless digital twins using commodity smartphone measurements and environmental data. Three specialized agents iteratively refine environment geometry, base station configuration, and material properties by minimizing discrepancies between measured RSRP and ray-tracing predictions. Real-world experiments show RSRP prediction error reduced from 11.46 to 5.25 dB MAE with cross-device transferability demonstrated.

Contribution: First end-to-end agentic system for autonomous WDT construction requiring minimal manual effort, using only smartphone RSRP measurements and publicly available environmental data to achieve high-fidelity ray-tracing calibration.

Relevance to your work: Directly relevant as high-fidelity digital twins are essential infrastructure for developing and testing AI-native air interfaces, particularly for training neural receivers and validating learned PHY components in realistic propagation environments.

A Token Service Interface for AI-Native RANs

Jianan Zhang, Peng Li, Xiang Cheng · arXiv cs.NI · 2026-10-02

★★★★■ This addresses a critical gap in task-oriented communication for 6G by proposing a concrete interface between AI services and RANs, with credible demonstration of joint radio-compute scheduling for emerging generative AI workloads.

The paper introduces a Token Service Interface (TSI) for AI-native RANs that allows generative and embodied AI applications to communicate token-stream requirements (service, temporal, and stateful semantics) to the network. TSI binds delivery requirements, readiness forecasts, and execution-state references to schedulable token sets, enabling importance-aware radio resource allocation and selective state migration. A drone inspection case study demonstrates how TSI coordinates RAN scheduling with edge computing to balance latency, throughput, and state-migration overhead.

Contribution: The novel contribution is an application-network interface abstraction that exposes token-level semantics (timing, importance, execution context) rather than byte streams, enabling RANs to co-optimize radio scheduling with AI inference and control loop dynamics.

Relevance to your work: Directly relevant to your work on AI-native air interfaces and semantic communication, as TSI exemplifies how PHY/MAC scheduling can exploit application semantics (token importance, timing) rather than treating traffic as opaque flows.

Cross-Site Transfer and Spatial Forgetting in OTA-Trained Neural Receivers

Riku Luostari, Dani Korpi, Harri Holma, Olav Tirkkonen · arXiv eess.SP · 2026-10-01

★★★★■ Addresses a critical gap between lab-trained neural receivers and real deployment by providing empirical evidence on generalization limits and adaptation costs using a rigorous multi-site OTA dataset, directly relevant to practical AI-native PHY deployment.

This paper evaluates cross-site generalization of neural receivers trained on over-the-air (OTA) measurements from a 5.88 GHz campaign spanning eight sites and four environment types. Four training regimes are compared: simulation-only, other-site data, joint multi-site training, and finetuning on deployment site after pre-training elsewhere. Results show that training on real OTA data from other sites yields 0.3 dB sensitivity gain over simulation at BER 10^-1, while finetuning on the deployment site provides an additional 0.3 dB gain but causes ~0.1 dB spatial forgetting on other sites.

Contribution: First systematic study quantifying cross-site transfer learning and catastrophic forgetting in neural receivers trained on real multi-site OTA measurements, demonstrating that other-site training provides meaningful but modest gains and that finetuning trades deployment-site performance for spatial forgetting.

Relevance to your work: Directly informs your AI-native air interface work by quantifying the real-world generalization challenges of learned PHY components and the trade-offs between site-specific adaptation and spatial robustness in neural receiver deployment.

ISAC (sensing + communication)

Learning-Based Beam Adaptation for Active/Passive-Aware Monopulse ISAC with Leakage Suppression

Jafar Norolahi, Alireza Vahid · arXiv eess.SP · 2026-10-02

★★★■■ Solid incremental work addressing a practical ISAC privacy/coexistence concern (leakage suppression) with a well-motivated RL formulation, though no deployment-scale validation or comparison to analytic baselines.

The paper develops a deep RL (Soft Actor-Critic) approach to adapt transmit beams in a dual-function radar-communication system that distinguishes cooperative (active) from non-cooperative (passive) targets using fused radar detections and uplink spatial evidence, while enforcing spatial leakage constraints to limit unintended illumination. The SAC agent optimizes a multi-objective reward balancing communication throughput, sensing accuracy, and leakage suppression across protected angular zones. Simulations show robust active/passive classification and the highest sum-rate among baselines when scaling from 1 to 12 passive targets under leakage budgets.

Contribution: First learning-based beam controller for ISAC that explicitly distinguishes active users from passive targets and enforces spatial leakage budgets via deep RL multi-objective optimization.

Relevance to your work: Directly relevant to your ISAC research agenda; demonstrates how ML can handle multi-objective beam management (sensing + comms + regulatory constraints) that analytic methods struggle with, and the active/passive fusion concept applies to real-world mixed-user scenarios.

Joint Discrete Pinching-Antenna Activation and Beamforming Design for Efficient PASS-ISAC

Joshua Ogundiya, Animesh Yadav, Octavia Dobre · arXiv eess.SP · 2026-10-02

★★★■■ Solid incremental work bridging the gap between theoretical PASS-ISAC designs and practical hardware constraints, though limited to specific multi-waveguide architecture without experimental validation.

This work addresses practical pinching-antenna systems (PASS) for ISAC by formulating a discrete antenna activation problem where transmit and receive antennas are selected from predefined locations rather than placed continuously. The authors develop JDPABO, an alternating optimization algorithm combining successive convex approximation and reweighted ■1 minimization to jointly optimize beamforming and sparse antenna activation under power and sensing SNR constraints. Simulations demonstrate near-continuous-placement performance at low sensing SNR requirements while achieving sparse activation.

Contribution: First treatment of discrete antenna activation for PASS-ISAC systems, replacing idealized continuous placement with practical selection from fixed hardware locations through a tractable joint optimization framework.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for understanding practical antenna selection and beamforming co-design trade-offs in joint sensing-communication systems with hardware constraints.